$$U = \text{unshuffle}([ZW_e; m; \ldots; m]) + P_d$$

$$W_e \in \mathbb{R}^{1024\times512},\ U \in \mathbb{R}^{197\times512}$$

$$\widehat{X}_p = [g_\phi(U)W_o]_{1:N}, \quad \widehat{X}_p \in \mathbb{R}^{196\times768}$$